

Improving Precision in Small Area Proportion Estimation Using Logit Transformation: A Case of Internet Utilization in Papua's Regencies, Indonesia (2021)

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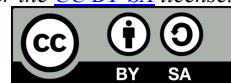
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ABSTRACT

The development of technology and communication reflects economic growth, with internet usage serving as one key indicator. While this indicator is generally available at the national or provincial level, reliable estimates at the regency level remain limited. Small Area Estimation (SAE) methods can address this gap by integrating survey data with census or administrative records. However, the basic SAE model may be less suitable for proportions, particularly in rare cases, due to violations of the normality assumption. This study shows that applying a logit transformation within the SAE framework improves the precision of proportion estimates. Using internet usage in Papua, Indonesia, as a case study, the results demonstrate that the logit-transformed SAE model outperforms both direct survey estimates and the basic SAE model.

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1. INTRODUCTION

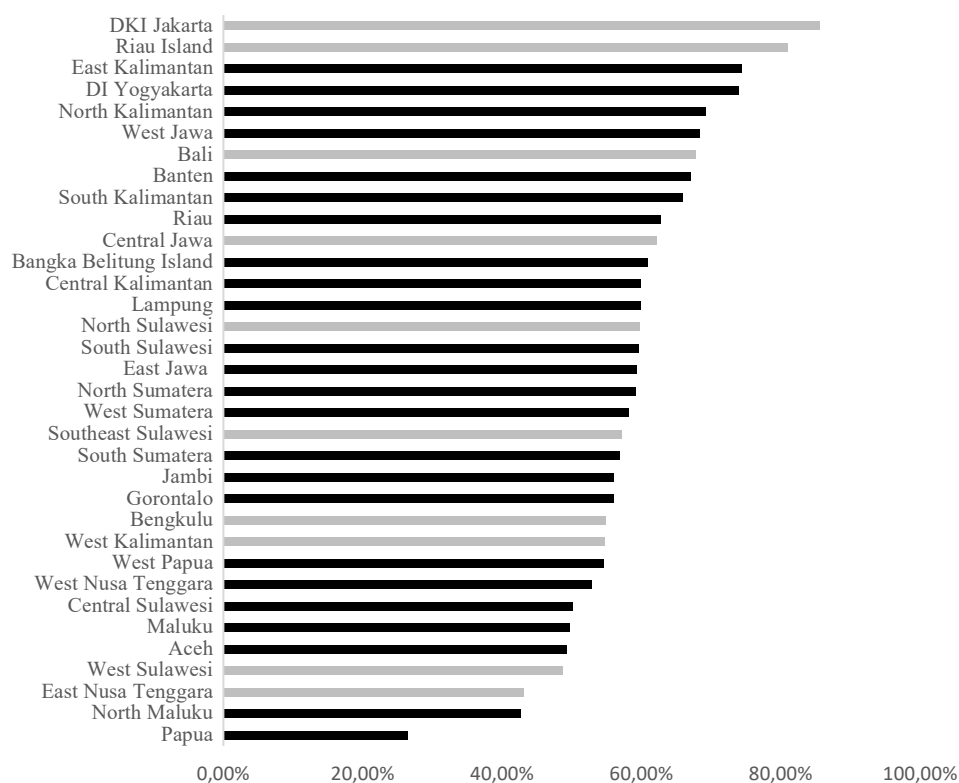
It cannot be denied that technological developments in a region can influence economic development. For example, Kallal *et al.* [1], showed that the diffusion of technology, information, and communication (ICT) has an impact on Tunisia's long-term economic growth. Akram *et al.* [2], also found that internet users, mobile phone users, and internet server GDP growth. Kurniawati [3], also shows that the penetration of telephone lines and cell phones is very capable of driving economic growth in middle-income countries in Asia.

Indonesia, one of the developing countries in Asia, is no exception and feels the positive impact of ICT on GDP per capita growth [4]. This has become a concern for the Indonesian government. Therefore, stated in one of its medium-term development plans for 2020-2024, "accelerating transformation in building and developing Technology, Information and Communication (ICT) infrastructure for equal distribution of broadband access are related to and connectivity throughout Indonesia" [5]. The acceleration of digital transformation can be monitored using the Technology, Information, and Communication Development Index (ICT Index) released by BPS-Statistics Indonesia [6]. The ICT Index is a composite index composed of three sub-indices. First, the ICT access and infrastructure subindex describes ICT readiness in terms of access and infrastructure. Second, the ICT use subindex describes ICT intensity in terms of the use of ICT. Finally, the ICT skills subindex describes ICT skills or expertise.

One of the indicators that make up the ICT index is "the proportion of the population using the internet". These indicators are generally only available at the national and provincial levels, so

only a few districts/cities can provide these indicators at the district/city level. On the other hand, the need for statistical data at the district/city level or even lower is increasing along with regional autonomy. As we know, the existence of regional autonomy gives authority to all regional governments to make policies in their respective regions. For this reason, local governments need supporting data in making their policies. Therefore, the availability of complete statistical data at this level is very important.

Figure 1 shows an overview of the "percentage of population using the internet" indicator according to provinces in Indonesia in 2021 and its availability at the district/city level. As shown in the picture, it can be seen that this indicator is available at the province level in Indonesia. Nevertheless, the dark color in the bar chart shows that this indicator is not yet available at the district/city level. The figure shows that only 10 out of 34, or 30 percent of provinces, have provided this indicator at the district/city level. The remaining 70 percent have not. This is our main concern, and Papua Province is one of them.



Source: BPS-Statistics Indonesia [6].

Figure 1. The percentage of the population using the internet by provinces in Indonesia in 2021 and availability to regency/city level (the black color indicates that data is not available at the district level)

Papua Province is one of the provinces in Indonesia with a low population density, approximately 14 people/km². The topography of this area consists mostly of forests, even the largest in Indonesia. This is one of the reasons why the percentage of the population using the internet in this province is the lowest (see Figure 1). Certainly, the low provincial estimation may impact disaggregation calculations to lower levels (i.e., regency level or below). The sample used to calculate the provincial level is too small if used for regency level estimates. This is a challenge in itself. According to Rao and Molina [7], this includes a "small area" framework. Therefore, this research aims to apply a small area model-based methodology to obtain precise estimates of the proportion of people using the internet at the regency level in Papua Province in Indonesia.

Data from the Indonesian National Socio-Economic Survey (SUSENAS) carried out by BPS-Statistics Indonesia can be used to calculate the proportion of the population using the internet by province. However, the survey design does not allow for estimates at lower levels. If it is forced

to be calculated at the district/city level, for example, the resulting estimation values generally have low precision.

This requires a sophisticated statistical method. One statistical method that can be used is the Small Area Estimation (SAE) model. As Rao and Molina [7], say, a small area estimation model is a valid and cost-effective alternative to direct estimates obtained from surveys aimed at obtaining estimates of population parameters when we have a small sample size. One of the most frequently used is the basic SAE Fay Herriot (FH) model. This model is applied to continuous or numeric data and requires the assumption of normality. For example, Ubaidillah *et al.* [8], used the multivariate SAE FH to estimate household per capita expenditure for food and non-food. Feriyanto *et al.* [9], used SAE FH to estimate this indicator, but they used imagery satellite as an auxiliary variable. Meanwhile, Sumarni *et al.* [10], relaxed the normality assumption by using a *t*-distribution for handling outliers.

However, sometimes this model cannot produce good estimates when the small area parameters are proportions, especially in rare cases. The basic SAE FH model commonly failed to reduce the standard error of the direct estimate. In addition, sometimes there is a normality assumption violation. The normality assumption should be met because hypothetical testing on model parameter estimates needs this assumption. To overcome this, Seeskin *et al.* [11], provided a solution by applying the logit transformation to the basic SAE FH model for the case of calculating the two-category proportion estimator. However, the mean square error estimates (MSE) were calculated simply using the Taylor series approach. Unfortunately, there is no analytical proof of the formula, and there is a possibility of estimation bias.

Besides that, Esteban *et al.* [12], also proposed a logit transformation to the basic SAE FH model for multicategory proportion estimation. The calculation of the MSE estimates was carried out using the bootstrapping approach. The advantage of bootstrapping is that it can produce a bias-corrected MSE estimator and does not require an analytical calculation [13]. However, our case only consists of two categories. Therefore, this study modifies the Esteban *et al.* [12], model for the case of two-category proportions.

2. RESEARCH METHOD

This study employs a model-based small area estimation framework to generate reliable estimates of internet utilization at the regency level in Papua Province. The analysis begins with direct estimation from the March 2021 National Socio-Economic Survey (SUSENAS), with precision evaluated using the relative standard error (RSE). To improve estimation reliability, the Fay-Herriot (FH) area-level model is applied by incorporating auxiliary variables derived from the 2021 Village Potential Statistics (PODES), such as electricity usage, educational infrastructure, and internet signal coverage, selected through multicollinearity tests and the Akaike Information Criterion (AIC). Since the parameter of interest is a proportion, a logit transformation is introduced to the FH model to address normality assumption issues and enhance stability, particularly for rare cases. Estimation is carried out using the Empirical Best Linear Unbiased Prediction (EBLUP), while a parametric bootstrap approach is employed to obtain bias-corrected mean square error (MSE) estimates. All computations are conducted in R using the *survey*, *sae*, and *sae.prop* packages.

2.1. Direct estimation

A direct estimate of the proportion of the population using the internet for each district level in Papua Province, the proportion at *i*th area ($\hat{p}_i$) can be calculated from March SUSENAS by formula (1).

$$\hat{p}_i^{DIR} = \frac{\sum_{j=1}^{n_i} W_{ij} Y_{ij}}{\sum_{j=1}^{n_i} W_{ij}} \quad (1)$$

where Y_{ij} is a binary data, $Y_{ij} = 1$ if the *j*th individual aged 5 years and over in the *i*th area used the internet and $Y_{ij} = 0$ if not. Meanwhile, W_{ij} is a sampling weight according to the survey design for each individual, n_i is the total of individual units in area *i*-th.

The Mean Square Error (MSE) of the direct estimator uses the following equation formula,

$$mse(\hat{p}_i^{DIR}) = \frac{\sum_{j=1}^{n_i} W_{ij}(W_{ij} - 1)(Y_{ij} - \hat{p}_i)^2}{\left(\sum_{j=1}^{n_i} W_{ij}\right)^2}. \quad (2)$$

2.2. Small area estimation

According to Rao and Molina [7], indirect estimation is a method of estimating by borrowing strength from sample data that not only come from its domain but also come from other areas through some auxiliary variables that do not contain errors, such as census data or administrative data. In the context of small areas, direct estimation leads to very large standard errors for subdomains where the sample size is too small. This makes indirect estimation one of the solutions to increase the effective sample size, thereby reducing the standard error. This method is known as Small Area Estimation (SAE). The SAE model consists of two basic models, namely area-level models and unit-level models.

SAE FH - basic model

This research focuses on area-level models. The basic area-level model was first introduced by Fay-Herriot in 1979. Suppose we have m small areas, a SAE Fay Herriot (SAE FH) model can be written as follows,

$$\hat{\theta}_i = \mathbf{x}_i^T \boldsymbol{\beta} + v_i + e_i, \quad i = 1, \dots, m \quad (3)$$

where θ_i is a small area parameter of i th area, $\hat{\theta}_i$ is a direct estimate of i th small area parameter from survey data. For example, if we use this model to estimate parameters in the form of proportions, $\hat{\theta}_i$ can be replaced by $\hat{p}_i^{DIR}$. $\mathbf{x}_i^T = (x_{1i}, x_{2i}, \dots, x_{li})^T$ is a vector of l auxiliary variables, $\boldsymbol{\beta}$ is a vector of regression parameters (fixed effect), v_i is a random effect for i th area where $v_i \sim N(0, \sigma_v^2)$, and e_i is an error model where $e_i \sim N(0, \Psi_i)$. In this area-level approach, Ψ_i is considered known and can be expressed as the sampling variance of the direct estimator (such as $\Psi_i = mse(\hat{p}_i^{DIR})$).

One estimation method that can be used in the SAE model is Empirical Best Linear Unbiased Prediction (EBLUP). The following is the EBLUP of the SAE FH model,

$$\begin{aligned} \hat{\theta}_i^{EBLUP} &= \hat{\gamma}_i \hat{\theta}_i + (1 - \hat{\gamma}_i) \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\beta}} &= \left[\sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^T / (\psi_i + \hat{\sigma}_v^2) \right]^{-1} \left[\sum_{i=1}^m \mathbf{x}_i \hat{\theta}_i / (\psi_i + \hat{\sigma}_v^2) \right] \\ \hat{\gamma}_i &= \frac{\hat{\sigma}_v^2}{\hat{\sigma}_v^2 + \Psi_i} \end{aligned} \quad (4)$$

where $\hat{\sigma}_v^2$ is the estimator of the random effect variance σ_v^2 . Estimation of σ_v^2 can be done using several methods, such as the moment method, maximum likelihood (ML) method, or restricted maximum likelihood (REML). More details about these estimates can be seen in Rao and Molina [7].

If we use REML, the MSE of the EBLUP estimator can be written as,

$$mse(\hat{\theta}_i^{EBLUP}) = g_{1i}(\hat{\sigma}_v^2) + g_{2i}(\hat{\sigma}_v^2) + 2g_{3i}(\hat{\sigma}_v^2) \quad (5)$$

$$\begin{aligned} g_{1i}(\hat{\sigma}_v^2) &= \frac{\hat{\sigma}_v^2 \Psi_i}{(\hat{\sigma}_v^2 + \Psi_i)} = \hat{\gamma}_i \Psi_i \\ g_{2i}(\hat{\sigma}_v^2) &= (1 - \hat{\gamma}_i)^2 \mathbf{x}_i^T \left[\frac{\sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^T}{(\hat{\sigma}_v^2 + \Psi_i)} \right]^{-1} \mathbf{x}_i \\ g_{3i}(\hat{\sigma}_v^2) &= [\Psi_i^2 (\Psi_i + \hat{\sigma}_v^2)^{-4}] (\hat{\theta}_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}})^2 \bar{V}(\hat{\sigma}_v^2) \end{aligned}$$

where $\bar{V}(\hat{\sigma}_v^2)$ is the asymptotic variance of the REML estimator $\hat{\sigma}_v^2$,

$$\bar{V}_{\hat{\sigma}_v^2} = 2 \left[\sum_{i=1}^m \frac{1}{(\psi_i + \hat{\sigma}_v^2)^2} \right]^{-1}.$$

SAE FH logit transformation

The SAE FH basic model is more suitable for normally distributed data. When the data is in the form of proportions or percentages, sometimes the normality assumption is not met. Esteban *et al.* [12] have proposed a new model that modifies the SAE FH basic model by replacing the dependent variable in model (3) with a logit transformation. This is a trivariate model (TFH), which can estimate the proportion of more than 2 categories. Meanwhile, in this study, there were 2 categories. Therefore, we adjusted the model, and we call it a logit transformation model. Based on this model, it is assumed that the proportion parameter is the inverse of the logit transformation,

$$p_i = \frac{\exp(\theta_i^*)}{1 + \exp(\theta_i^*)} \quad (6)$$

where $\theta_i^* = \mathbf{x}_i^T \boldsymbol{\beta} + v_i^*$, $v_i^* \sim N(0, \sigma_v^{2*})$

Therefore, the SAE FH transformation logit model can be expressed as,

$$\hat{\theta}_i^* = \log \left(\frac{\hat{p}_i^{DIR}}{1 - \hat{p}_i^{DIR}} \right) = \mathbf{x}_i^T \boldsymbol{\beta} + v_i^* + e_i^*, \quad e_i^* \sim N(0, \Psi_i^*) \quad (7)$$

where, $\hat{p}_i^{DIR}$ is the direct estimate of the proportion from the survey.

From this model, in the same way as the SAE FH basic model, we get a new EBLUP, $\hat{\theta}_i^{EBLUP*}$. To get EBLUP from the SAE FH logit transformation model, we first need to modify the sampling error variance of the proportion from the survey. According to Esteban *et al.* [12], the variance of the logit transformation of a direct estimate of the proportions can be approximated by,

$$\begin{aligned} \Psi_i^* &= \mathbf{H} \text{var}(\hat{p}_i^{DIR}) \mathbf{H}^T \\ \mathbf{H} &= q(\mathbf{I}_{q-1} + \mathbf{1}_{q-1} \mathbf{1}_{q-1}') \end{aligned} \quad (8)$$

where $\mathbf{I}_a$ is the $a \times a$ identity matrix, $\mathbf{1}_a$ is the $a \times 1$ vector with all the components equal to one, and q is the number of categories. In binary data, $q = 2$. If we assume that the direct estimator is unbiased, then the sampling error variance is the same as the MSE (2), so

$$\text{var}(\hat{p}_i^{DIR}) = \text{mse}(\hat{p}_i^{DIR})$$

Furthermore, the estimated value of the proportion parameter by the SAE FH logit transformation can be expressed as follows,

$$\hat{p}_i^{EBLUP} = \frac{\exp[\hat{\theta}_i^{EBLUP*}]}{1 + \exp[\hat{\theta}_i^{EBLUP*}]} \quad (9)$$

where $\hat{\theta}_i^{EBLUP*}$ is the EBLUP resulting from the logit transformation of the proportion as in equation (7). The calculation to obtain this EBLUP is the same as the EBLUP of the SAE FH model (4).

The estimation of the MSEs of the proportion estimation can be computed by the parametric bootstrap. The steps are as follows,

1. Fit the SAE logit transformation model (7) and estimate the beta coefficient ($\hat{\beta}$) and variance random effect variance area ($\hat{\sigma}_v^{2*}$).
2. Do bootstrap as many as B iterations ($b = 1, \dots, B$):
 - 1). Generate $\theta_i^{*(b)}$ as a subsample parameter from each area, and get the parameters bootstrap proportion $p_i^{*(b)}$.

$$\theta_i^{*(b)} \sim N(\mathbf{x}_i^T \hat{\beta}, \hat{\sigma}_v^{2*})$$

$$p_i^{*(b)} = \frac{\exp[\theta_i^{*(b)}]}{1 + \exp[\theta_i^{*(b)}]}$$

- 2). Generate new data $\hat{\theta}_i^{*(b)}$

$$\hat{\theta}_i^{*(b)} \sim N(\theta_i^{*(b)}, \Psi_i^*)$$

- 3). Calculate the EBLUP ($\hat{\theta}_i^{EBLUP*(b)}$) based on the data $(\hat{\theta}_i^{*(b)}, \mathbf{x}_i)$ with variance sampling error (Ψ_i^*).
- 4). Calculate the logit transformation EBLUP of the proportion bootstrap.

$$\hat{p}_i^{EBLUP(b)} = \frac{\exp[\hat{\theta}_i^{EBLUP*(b)}]}{1 + \exp[\hat{\theta}_i^{EBLUP*(b)}]}$$

3. Calculate the bootstrap MSE estimator of $\hat{p}_i^{EBLUP}$:

$$mse_B(\hat{p}_i^{EBLUP}) = \frac{1}{B} \sum_{b=1}^B (\hat{p}_i^{EBLUP(b)} - p_i^{(b)})^2 \quad (10)$$

2.3 Relative standard error (RSE)

The reliability of an estimator can be seen from the MSE. However, if we compare several methods, we should use standard measurements. The relative standard error (RSE) is a standard measure of precision. The RSE of an estimator $\hat{\phi}_i$ can be calculated as follows,

$$RSE(\hat{\phi}_i) = \frac{\sqrt{mse(\hat{\phi}_i)}}{\hat{\phi}_i} \times 100\% \quad (11)$$

We interpreted this measure according to BPS-Statistics Indonesia [14]. An estimator is said to have good precision if the RSE value is less than 25%. If the value is between 25% and 50%, it indicates that caution is needed in using the estimation results, and if the value is greater than 50%, the estimation results are not good and should not be used.

2.4 Data sources

A person is said to use the internet if he takes the time to use the internet so that he can utilize or enjoy internet facilities such as searching for literature/references, searching/sending information/news, communication, email, chatting, social media, online games, and so on [14]. In Indonesia, data on the proportion of the population using the Internet can be reached from the National Socioeconomic Survey (SUSENAS), which targets the population aged 5 years and over. This survey is carried out every semester, namely in March and September, to collect official socio-economic statistical data. The number of samples in March was greater than in September. It is about 300,000 households sampled in all regions of Indonesia.

Several indicators of sustainable development goals (SDGs) at the district level can be estimated well from the March SUSENAS. However, some do not, especially in areas with rare cases, such as estimating the proportion of the population using the internet in Papua at the district level. The sample size is not enough. If estimates are forced, the results will not be reliable. However, we can still use the direct estimator of the proportion from this survey as a response variable in the SAE model. We have 29 districts in Papua Province in 2021 as small areas studied.

In addition, to obtain small area statistics based on the SAE model, census data or administrative data that is free from sampling error is required as auxiliary variables (or independent variables in the regression model). Therefore, Village Potential data (PODES 2021) can be used as a source for obtaining these auxiliary variables. PODES data is an administrative data collection at the village level, which generally consists of village infrastructure data.

The following is some literature that reveals what variables may be related to the variable of interest. Xiaoming and Kay [15], stated that Internet penetration is related to a country's wealth, telecommunication infrastructure, and urbanization. Mocnik and Sirec [16], show that the strongest positive and most significant impact on Internet use was the ICT infrastructure and people's capabilities, followed by income distribution. Moreover, Turkan [17], emphasized that per capita Gross Domestic Product (as an area welfare symbol) is related to electricity consumption. Atkin [18] also showed that income was related to cable subscribership. So we can use the electricity users variable to approximate the income variable. On the other hand, Krug et al. [19], found that the level of education can have an impact on internet usage.

To make indirect predictions, the selection of auxiliary variables is first carried out. In this study, there are 18 candidate auxiliary variables from PODES data, which include variables on the availability of electricity, the availability of educational infrastructure, the availability of economic infrastructure, and internet access, such as ownership of mobile phones/telephones and BTS towers, etc. In this research, aggregation of PODES data was carried out at the regency level.

Then, several variables are selected that meet the requirements of non-multicollinearity using a Variance Inflation Factor (VIF) value of less than 10. Furthermore, the selected variables are used in the basic regression model. First, a basic regression model is built from all selected variables. If there are variables that are not significant, then several new models are formed by combining several significant variables. The variables that build the model with the lowest Akaike Information Criterion (AIC) value of all models formed by backward elimination are the selected variables for further use in SAE modeling.

Finally, we got three auxiliary variables from PODES data that were used to predict the proportion estimated value of the population using the internet at the regency level in Papua Province. These variables are the ratio of the number of families using PLN electricity to the total population (ELECTRICITY), the average number of senior high schools/equivalent at each village/ward (EDUCATION), and the proportion of villages/wards where most of their areas have cellular phone internet signal/mobile 3G/H/H+/EVDO (INTERNET_SIGNAL).

2.5 Software

Summary statistics and simple visualizations are calculated using MS Excel. We use R software (or R Studio) for more advanced calculations. Direct estimates are calculated using the "*survey*" package [20], basic SAE FH modeling using the "*sae*" package [21], and logit transformation SAE FH modeling using the "*sae.prop*" package [22].

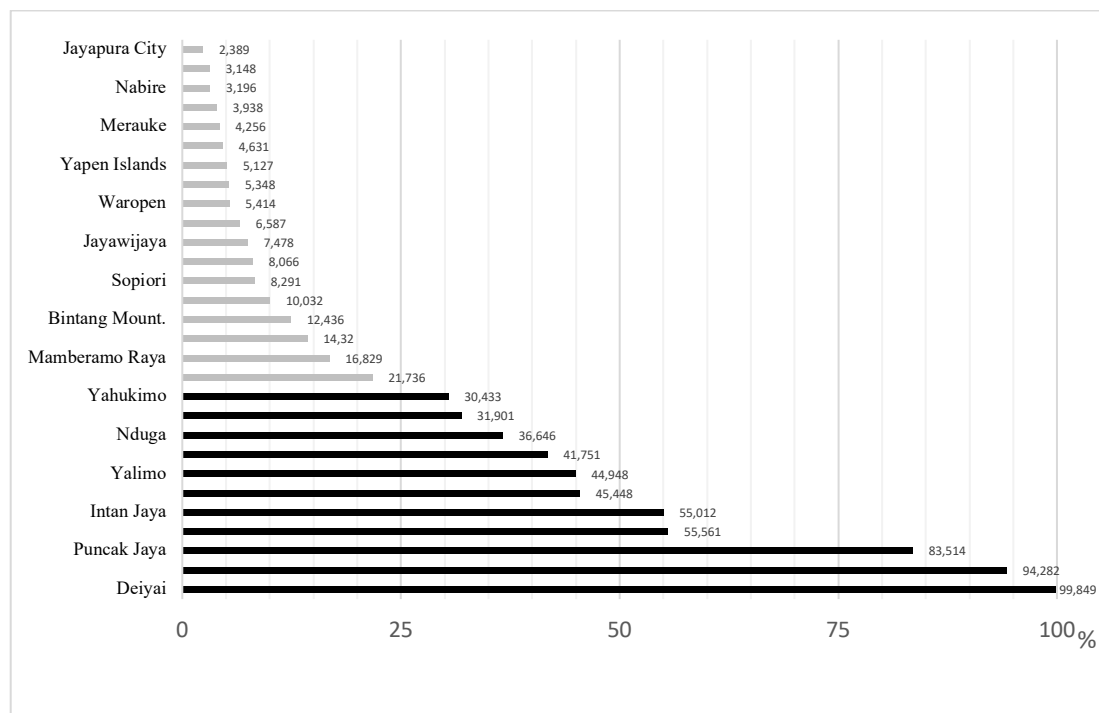
3. RESULTS AND DISCUSSION

This section presents the findings of the study along with comprehensive interpretations. The results are structured to first describe the reliability of direct estimates of internet utilization at the regency level in Papua Province, followed by the outcomes of the small area estimation (SAE) models. Comparisons are made between direct estimates, the basic Fay-Herriot (FH) model, and the logit-transformed FH model to evaluate their relative performance in terms of precision, as measured by the relative standard error (RSE). Furthermore, the discussion highlights the contribution of selected auxiliary variables, examines the reduction in estimation errors achieved

by the proposed approach, and illustrates spatial disparities in internet usage across regencies through mapping.

3.1. Direct estimate of the proportion of the population using the internet at the regency level in Papua Province in 2021: an overview

The RSE values generated from direct estimation can be seen in Figure 2. This figure depicts eleven of the twenty-nine districts/cities with RSE values of more than 25 percent, marked with a black bar chart. We see that starting from the Yahukimo district to the Deiyai district, the RSE values range from 30.44 percent to almost 100 percent. The maximum value occurs in the Deiyai district with an RSE of 99.849 percent. This shows that many direct estimates in this case are unreliable, so we must be careful when using this direct estimate.



Source: SUSENAS March 2021 (own construction)

Figure 2. RSE of direct estimate by district/city in Papua Province (in %)

This low precision is due to the insufficient number of SUSENAS March samples to estimate the indicator of the proportion of the population using the internet at the district level in Papua Province. Therefore, the SAE method is needed to produce better precision estimators. We tried two SAE models, namely the basic SAE FH model and the SAE FH logit transformation model, and then compared them to obtain the best model.

3.2. Small area estimation of the proportion of the population using the internet at the regency level in Papua Province in 2021

The regression parameter estimates of both the SAE FH basic model and the SAE FH logit transformation model can be seen in Table 1. All of the auxiliary variables are positively significant on the variable of interest with a significance level of 0.01 and 0.05.

The variable of internet signal coverage approximated by the proportion of villages/wards where most of their areas have cellular phone internet signal/mobile (INTERNET_SIGNAL) variable has the greatest effect among the three variables. Followed by the variable of senior high school education infrastructure (EDUCATION). The magnitude of this effect value indicates the magnitude of the contribution of the variable in calculating the estimated proportion of the population using the Internet. The greater the effect, the greater the contribution.

The impact of electricity users is also significant. All of these variables have a positive effect on predicting the proportion of the population using the internet in Papua. It means that an increase in ICT infrastructure, education, and access to electricity can lead to an increase in the proportion of the population using the internet in that area.

Table 1. Regression parameter estimates of the SAE FH basic model and the SAE FH logit transformation model

	Coefficient	Std. error	t-value	p-value
SAE FH basic model:				
Intercept	0.013	0.028	0.466	0.641
ELECTRICITY	0.185	0.051	3.628	0.003**
EDUCATION	0.473	0.077	6.191	0.000**
INTERNET_SIGNAL	1.345	0.397	3.383	0.007**
SAE FH logit transformation model:				
Intercept	-4.353	0.478	-9.107	0.000**
ELECTRICITY	1.822	0.841	2.167	0.030*
EDUCATION	3.612	1.261	2.864	0.004**
INTERNET_SIGNAL	15.087	6.517	2.315	0.021*

Note: ** significant level 0.01, * significant level 0.05.

The estimation results based on the basic SAE-FH model and the SAE-FH logit transformation model are summarized in Table 2. From Table 2, it can be seen that the estimates from both the basic SAE-FH model and the SAE-FH logit transformation are very similar to the direct estimates. The relative standard errors (RSE) of the basic SAE-FH model are also not significantly different from those of the direct estimates. In contrast, the estimates obtained using the SAE-FH logit transformation model appear better, as indicated by a reduction in RSE values, although some RSEs remain above 25%.

Table 2. Summary statistics of the estimated results by three methods (direct estimation, the basic SAE FH model, and the SAE FH logit transformation model)

Statistics	Direct Estimation		SAE FH Basic		SAE FH Logit Transformation	
	Proportion estimates	RSE (%)	Proportion estimates	RSE (%)	Proportion estimates	RSE (%)
Minimum	0.001	2.39	0.001	2.38	0.001	1.40
1st quartile (Q1)	0.008	5.24	0.008	5.21	0.009	3.11
Median	0.097	12.44	0.098	12.45	0.097	4.89
Mean	0.199	26.29	0.198	26.06	0.199	8.74
3rd quartile (Q3)	0.377	43.35	0.377	42.74	0.364	12.37
Maximum	0.704	99.85	0.707	99.02	0.705	32.56

3.3. Model evaluation

Figure 3 depicts a drastic decline in RSE values using SAE FH logit transformation compared to direct estimates and basic SAE FH models. Meanwhile, the direct estimation results do not appear to have been enhanced by the basic SAE FH method.

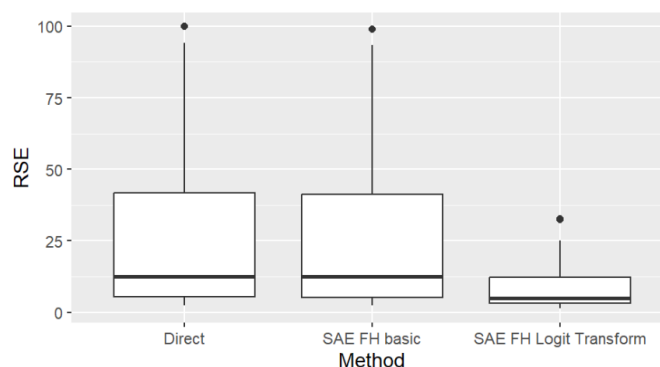


Figure 3. RSE comparison between direct estimation, the SAE FH basic, and the SAE FH logit transformation

Further details can be seen in Table 2, there is a reduction in the value of RSE. The average RSE results of the direct estimation and the SAE FH Basic method are about 26 percent, decreasing to 8.74 percent by the SAE FH method of logit transformation. Similarly, the reduction occurs at its maximum value, direct estimation, and basic SAE FH are about 99 percent to 32.56 percent by SAE FH logit transformation. Looking back at Figure 3 in more detail, the RSE of the SAE FH logit transformation estimates exceeding 25% appears in only one area (district).

In other words, based on the RSE values of the two SAE models, it can be inferred that the basic SAE FH model has not been able to improve the precision of the estimation. This means that there are still many unreliable estimates at the regency level by the SAE FH basic model. Meanwhile, the RSE produced by the logit transformation SAE FH model looks much smaller. Even so, when viewed from the estimation results, it is only slightly different. Therefore, the SAE FH logit transformation is a better method in this case.

3.4. Mapping Internet Usage Proportions at the Regency Level in Papua, Indonesia (2021) Using SAE-FH Logit Transformation

Figure 4 illustrates the estimation result of the SAE FH logit transformation for the proportion of the population using the internet in Papua Province, Indonesia 2021. The darker the color on the map, the lower the proportion value.

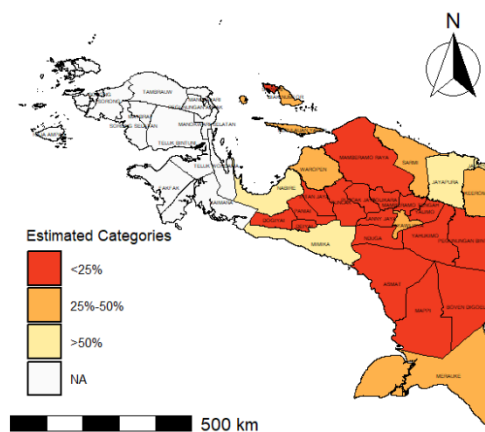


Figure 4. Regency-Level Internet Usage Proportion Estimates in Papua Province, Indonesia (2021) Using SAE-FH Logit Transformation (The NA value indicates that West Papua Province was excluded from the case study)

We can see from Figure 4 that the dominant color is dark and darker (red). This shows that the estimated proportion of the population using the internet generated at the district level in Papua Province generally tends to be low, which is less than 0.5. Based on Table 2 before, the average can be seen to be around 0.199. This can be interpreted that in almost all districts in Papua Province, the population still rarely uses the internet, which is only around 19.9%. Only Nabire, Jayapura, and Mimika Regencies, along with Jayapura City, show better conditions, where more than 50% of the population uses the internet.

The lowest proportion estimate value (< 0.25) is depicted in the darkest color. This occurs in 18 districts or more than 50 percent of the 29 districts, namely Deiyai Regency, Central Mamberamo, Mamberamo Raya, Intan Jaya, Puncak Jaya, Paniai, Lanny Jaya, Yalimo, Nduga, Dogiyai, Tolikara, Yahukimo, Puncak, Gunung Bintang, Asmat, Mappi, Supiori, and Boven Digoel Regency. These districts are generally inland areas, with many wild tribes living in the mountains and even in the forest. Furthermore, the lighter color shows the category of the proportion of the population using the internet between 0.25-0.50 in as many as 7 districts, namely, Jayawijaya, Waropen, Yapen Islands, Sarimi, Keerom, Merauke, and Biak Numfor.

The high disparity between regencies indicates an imbalance in ICT development in Papua Province in 2021. This can also be seen from the estimated proportion range value, with a

minimum result of 0.001, while the maximum is 0.705. In addition, there are 25 districts where the percentage of the population using the internet is still below 50 percent. This indicates that the development of internet infrastructure in Papua is not evenly distributed. As stated by the BPS-Statistics of Papua Province [23], the number of villages that have BTS towers is only around 12 percent of the 4,946 existing villages. The worst-case scenario occurs in forest and mountainous areas, where only around 1 percent, such as in Yahukimo Regency, the number of villages have BTS towers is only 3 villages out of 515 villages.

In addition, this problem could be the impact of the uneven development of human resources in Indonesia, especially in the Eastern Indonesia Region (KTI) [24]. This follows what was shown by the BPS-Statistics of Papua Province [25] that the mean years school (MYS) for people aged 15 years and over in Papua Province 2021 was 6.76, meaning that they could only complete elementary school education. Not only that, the fact is that more than 50 percent of districts, the MYS for less than 6 years. The lowest occurred in Nduga Regency, which was only 1.42 years.

4. CONCLUSION

It can be inferred from the research results and discussion that the SAE FH logit transformation model can decrease the relative standard error (RSE) values for proportion estimations. The SAE FH logit transformation model can produce a more precise estimate than direct estimation and the basic SAE FH model, especially in rare cases when estimating proportions for small areas. This method can reduce the RSE of all of the proportion estimates of the population using the internet at the regency level in Papua Province, Indonesia 2021, which previously had an RSE value of more than 25 percent in direct estimation and the basic SAE FH.

Thus, this study can be said to have succeeded in producing a reliable estimate of the proportion of the population using the internet at the regency level in Papua Province using the SAE FH logit transformation method. This can be used as information for further ICT development planning in Papua Province. Besides that, the map image makes it easier to capture the information.

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